

WATHEQ

Corporate Profile

Advanced Electronic Archiving & Document Management
System with High Compliance and Data Security

About The System

Watheq is a comprehensive electronic archiving and correspondence management solution. The system enables full digitization of transactions, workflow tracking, and paperless environments. Designed flexibly to meet corporate and government requirements, Watheq focuses heavily on high compliance, intelligent archiving capabilities, and exceptional search functionalities.

Core Competitive Features

Advanced Intelligent Search

Precise search within document metadata and even inside attachment texts leveraging advanced PostgreSQL GIN Indexes algorithms, providing instant results.

Content Extraction (OCR Engine)

A smart system powered by Tesseract to convert scanned PDFs and images into fully searchable text, indexing them in real-time seamlessly.

Document Lifecycle (Retention)

Automated processes to manage retaining, archiving, or disposing of expired documents securely, accompanied by auto-generating official Disposal Certificates.

Legal Hold

Instantly freeze and hold documents required for legal auditing or investigations, overriding their scheduled automated retention and disposal policies.

System Architecture

Watheq adopts modern software engineering by completely decoupling the user interfaces (Client Layer via React) from the processing layer (API Layer using Node.js). This ensures uncompromising security, immense scalability, and smooth continuous development. Heavy operations such as OCR are managed asynchronously via background task queues.

[System Flow Data & Architecture Diagram]

Backend Implementation

Technology	Primary Role in Ecosystem
Node.js & Express	High-performance RESTful API server for fast request handling
PostgreSQL & Prisma	Primary relational database paired with robust ORM bridging
Redis + BullMQ	Background job queuing orchestrator for OCR, notifications, and heavy loads
Tesseract.js & PDF-Parse	Text extraction processors feeding the intelligent search engine

Frontend Implementation

Technology	Primary Role in Ecosystem
React 18 & TypeScript	Building fast, type-safe, and highly responsive user interfaces
Tailwind CSS	Fluid stylistic implementation enabling robust responsive layouts
react-i18next	Comprehensive internationalization (RTL/LTR) natively built-in

WATHEQ

Your Gateway to a Secure Digital Transformation